**TABLE OF CONTENTS**


| CONTENTS | DESIGNATED PAGE NUMBER(S) |
|---|---|
| Small Business Concern | 1 |
| Technical Merit Summary Including Key Personnel | 2-4 |
| Concept & Significance of Problem/Opportunity | 5 |
| Defense Need | 6 |
| OUSD(R&E) Critical Technology Areas and/or Commercial Space Strategy Mission Area | 6 |
| Transition Plan | 7 |
| Summary of the Proposed Solution's Commercialization and Non-Defense Commercial Potential Business Plan | 8, 9 |
| Related Work (Dual-Use) | 10 |
| Non-Proprietary Work Plan Outline and Phase I Objectives | 11, 12 |
| Foreign Persons | 13 |
| Glossary | 14-15 |

**SMALL BUSINESS CONCERN (SBC):** Tychon AI - [redacted]
**Proposal Title:** Radiation-Tolerant Cognitive Control Loop Enabling Quantum Navigation in LEO & Cislunar Space
**Proposal Number:** FX255-PCS02-0044

**TECHNICAL MERIT SUMMARY**

Currently, our technology is in the **active research and early developement** stage.

**Scientific and Technical Baseline**

The U.S. Space Force (USSF) requires resilient Positioning, Navigation, and Timing (PNT) independent of GPS. Quantum sensors, specifically Nitrogen-Vacancy (NV) diamond magnetometers, offer a promising pathway for GPS-denied navigation. Recent research has demonstrated state-of-the-art room-temperature sensitivities for these sensors, reaching **18 pT · Hz$^{-½}$** (Stürner et al., *Phys. Rev. Applied*, 2021) and **9.4 pT · Hz$^{-½}$** without cryogenic assistance (Clevenson et al., *Appl. Phys. Lett.*, 2015).

However, achieving this performance in an operational environment is severely limited by **spin decoherence** which is the loss of the fragile quantum state required for measurement. The primary challenge is that the dominant noise sources responsible for decoherence in high-quality synthetic diamonds are dynamic and complex. These include:

1. **Internal Noise:** Fluctuations in the surrounding electron spin bath, primarily from substitutional nitrogen (P1 center) impurities.
2. **External Noise:** Magnetic and electric field noise from the diamond surface, along with thermal drifts experienced on-orbit.

The current baseline for controlling these sensors relies on classical digital signal processing (DSP) like PIDs and PLLs. These fixed-algorithm methods cannot adapt to the complex, time-varying noise signatures that degrade sensor performance, creating a critical capability gap.

**The Innovation: A Radiation-Tolerant, Hybrid AI Control Architecture**

We propose a novel, radiation-tolerant, hybrid "AI-in-the-loop" control architecture to overcome the limitations of classical DSP. Our solution directly addresses the challenge of spin decoherence by introducing machine learning into the real-time quantum control loop. This breakthrough is possible only now due to the convergence of two key technologies: the commercial availability of radiation-tolerant SoCs with dedicated AI accelerators and the maturation of advanced machine learning techniques for real-time signal processing.

**Hardware Platform: Microchip RT PolarFire SoC**

This platform is foundational to our innovation due to its unique suitability for space applications.

- **Radiation Tolerance:** The RT PolarFire's flash-based SONOS process technology provides inherent immunity to radiation-induced configuration Single-Event Upsets (SEUs), a catastrophic failure mode for competing SRAM-based FPGAs. This eliminates the need for complex and power-hungry mitigation circuitry, increasing system reliability. The device family has achieved MIL-STD-883 Class B and QML Class Q qualifications, with a clear path to QML Class V.
- **Power Efficiency:** The platform offers up to **50% lower total power consumption** compared to competing mid-range FPGAs, a critical advantage for SWaP (Size, Weight, and Power)-constrained missions.

**Core Innovation: The Hybrid AI Control Loop**

Our architecture integrates a fast classical loop with an AI-adaptive loop, creating a system that intelligently filters noise and adapts to environmental drift.

1. **Fast Classical Loop (<150 µs latency):** A high-bandwidth stabilization loop implemented in the FPGA fabric. This latency is highly defensible; the sum of modern ADC conversion (~100 ns) and parallel FPGA logic processing (tens of ns to a few µs) is well under this target. Similar real-time control systems have demonstrated latencies as low as 17.6 µs.
2. **AI-Adaptive Loop (<1 ms latency):** An INT8-quantized Convolutional Neural Network (CNN) that performs real-time quantum state discrimination and adaptive noise filtering. A CNN is required because the target noise sources exhibit complex, non-linear, and time-varying correlations that are intractable for traditional filtering algorithms like Kalman filters. This sub-millisecond latency is enabled by using the **VectorBlox SDK** to accelerate our quantized model on the PolarFire's specialized AI core, avoiding the need for a power-hungry GPU.

This hybrid approach allows the system to maintain high-bandwidth control while using the AI loop to continuously adapt to and cancel out the complex noise sources that limit classical systems. Specifically, the CNN will output corrective parameters to the classical control loop in real-time, adjusting its gains and setpoints to actively cancel the measured noise and stabilize the quantum

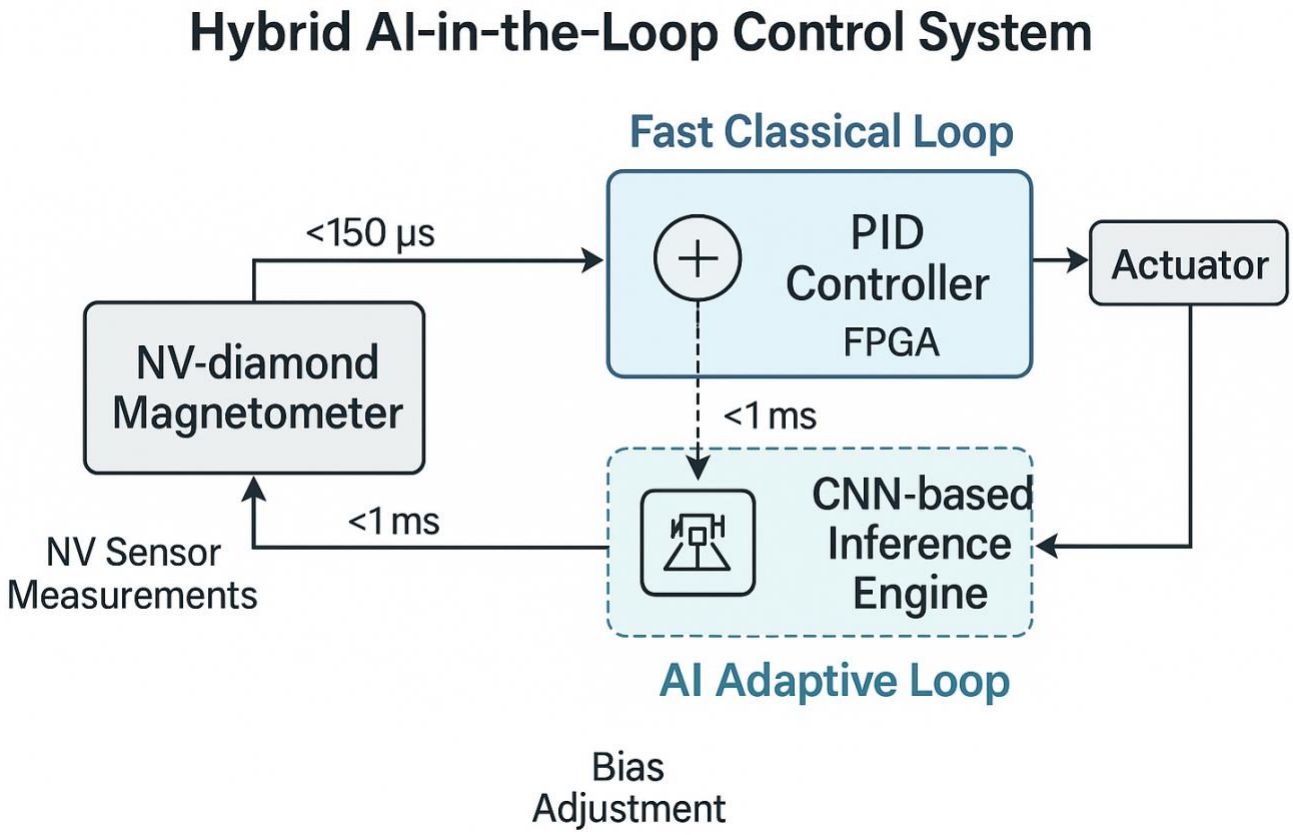
state. While this Phase I effort focuses on real-time magnetometer control, our Phase II roadmap integrates MEMS IMU, CSAC, and star-tracker data into the same SoC to create a complete, radiation-tolerant PNT solution.

**AI Validation and De-risking**

The most significant technical risk in any AI hardware project is the "synthetic-to-real" generalization gap, where a model trained on clean, simulated data fails when deployed on noisy, real-world hardware.

To mitigate this risk, our AI validation plan moves beyond using purely physics-based synthetic data. A dedicated task in our Phase I work plan is focused on **realistic noise characterization and data augmentation**. We will develop validated, stochastic models for the dominant noise sources previously identified and the magnetic noise from the P1 center spin bath including low-frequency thermal drift. These noise models will be used to corrupt the clean, physics-based data, generating a more realistic and challenging training dataset. Training our CNN on this augmented data will force it to learn to distinguish the underlying quantum signal from the noise, dramatically improving its robustness and its probability of successful transition to real hardware in Phase II.

**System Architecture and Implementation:** The proposed architecture (shown below) demonstrates the integration of the AI accelerator within the feedback loop, enabling real-time adaptation of the classical control parameters.



**Key Personnel:**

| NAME / ROLE | EDUCATION | EXPERIENCE AND/OR CAPABILITIES |
|---|---|---|
| Jason Phillip | B.S. Computer Science | Principal Investigator; AI/FPGA Control Architect; developed and deployed CNN-based star tracker inference engines on the Microchip RT PolarFire SoC; built physics-informed synthetic data pipelines analogous to NV-diamond synthetic data; designed and integrated hybrid classical/AI low-latency control loops for real-time sensor stabilization and adaptation; founder/operator of Tychon AI with government contract execution experience. |
| System Architect & Integration Lead | Ph.D., Mechanical Engineering | System Architect & Integration Lead; spacecraft navigation and control verification experience; led CubeSat-class system-level testing and modular PNT integration; provides SBIR-relevant technical leadership and transition planning. |
| FPGA & Embedded Systems Engineer | M.S. Computer Science; B.S. Electrical Engineering Technology | FPGA & Embedded Systems Engineer; implements real-time adaptive inference and stabilization logic; expert in low-latency digital signal processing on radiation-tolerant fabrics; integrates AI accelerators with control loops for robust on-board decision making. |
| Power Electronics & Integration Engineer | B.S. Electrical Engineering and M.S. Power and Energy Systems | Power Electronics & Integration Engineer; designs and stabilizes analog/power front ends for sensitive quantum sensor heads; experienced in defense/aerospace embedded systems with emphasis on thermal and noise robustness. |

## CONCEPT & SIGNIFICANCE OF PROBLEM/OPPORTUNITY

### The Operational Problem

U.S. Space Force (USSF) satellites operating in GPS-degraded or GPS-denied environments must maintain precise knowledge of their attitude, position, and time to ensure mission success. The classical control systems used for today's cutting-edge quantum sensors fail to meet the demands of a contested space domain, leading to direct and critical operational consequences:

- **Satellites Cannot Reliably Maneuver in Combat:** Classical control loops with high latency (>5 ms) cannot maintain attitude lock during the rapid slewing maneuvers required for threat evasion or retargeting. This creates significant **vulnerability windows** and degrades the pointing accuracy of critical payloads like laser communication terminals or directed energy systems.
- **A Single Upset Can Cause Total Mission Loss:** Current processors often rely on SRAM-based FPGAs, which require cumbersome Triple Module Redundancy (TMR) for radiation mitigation. TMR inflates Size, Weight, and Power (SWaP) and still carries a residual risk of unrecoverable configuration upsets (SEUs) in the Van Allen belts, which could **permanently disable the PNT system and lead to total mission loss**.
- **Constant Ground Intervention Reduces Mission Availability:** Static, non-adaptive control algorithms cannot compensate for performance drift from on-orbit thermal cycling and sensor aging. This forces frequent, time-consuming ground-commanded recalibrations, **reducing the satellite's availability** for its primary intelligence, surveillance, and reconnaissance (ISR) or communications mission.

### Proposed Solution

We will create a room-temperature, radiation-tolerant **"AI-in-the-loop" control engine** on a Microchip RT PolarFire SoC FPGA. A µs-class classical loop stabilizes the NV ensemble while an INT8-quantized CNN (accelerated with VectorBlox) provides µs-scale adaptive bias cancellation. Our AI-Cognitive Engine is specifically designed to mitigate these operational failures. It enhances maneuverability through low-latency control, ensures survival with radiation-hardened hardware, and maximizes mission uptime via real-time adaptive calibration.

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### Dual-Use Path & R/R&D Compliance

[Commercial strategy omitted for publication]

**R/R&D Compliance:** All tasks involve new algorithm development, mixed-signal FPGA design, and sensor characterization—no procurement-only activities—fully meeting SBIR R/R&D intent.

**DEFENSE NEED**

[Named customer and transition-target discussion omitted for publication]

**Comparison to User Baseline:** Compared with the baseline Honeywell HG4934SRS space IMU Honeywell Aerospace and Sinclair ST-16 star tracker Satsearch, our AI-Cognitive Engine delivers an *order-of-magnitude* better attitude hold while cutting SWaP in half. Integration with Microchip's radiation-tolerant CSAC Microchip further adds assured time. All components are already TRL 8-9 and Phase II focuses on algorithm fusion and environmental qualification, not sensor invention. Our approach de-risks the hardware platform by leveraging a suite of high-TRL (TRL 8-9) components, including the Honeywell space IMU, Sinclair star tracker, and Microchip CSAC. Our core R&D effort, currently at TRL 2, focuses on the novel AI-driven fusion algorithms and µs-scale control loops required to integrate these sensors into a single, cohesive PNT engine. This is work that addresses a critical gap not covered by sensor invention alone.

| **Proposed Improvement Metric** | **Baseline*** | **Proposed AI-Cognitive Engine** | **Operational Impact for USSF** |
|---|---|---|---|
| **Attitude Hold Accuracy** (*Heading Drift)* | >0.1°/hr | **<0.05°/hr** | Enables longer observation dwells for high-resolution ISR; improves pointing accuracy for tactical laser communications. |
| **Power Consumption** *(Total PNT System Power)* | ~25 W | **<12 W** | Frees >10 W for primary payloads (e.g., advanced transponders) or **extends mission life by over 30%** on a typical 12U bus. |
| **Radiation Resiliency** *(Config. Upset Rate, GEO)* | $10^{-5}$ upsets/bit-day (*SRAM-based)* | **<$10^{-10}$ upsets/bit-day** *(Flash-based)* | Enables long-duration, autonomous operations in high-radiation orbits (MEO/GEO/cislunar) with drastically reduced risk of mission-ending failures. |
| **Control Loop Latency** (*Sensor Update Rate)* | <200 Hz | **>1 kHz** | Allows the ADCS to track faster dynamic maneuvers, **improving survivability** against kinetic threats and enabling rapid retargeting. |

*Baseline (System with HG4934SRS IMU & SRAM FPGA Processor)

**Breadth of Applicability: A Platform-Level Enabler for Quantum Sensing**

Our hybrid control architecture is fundamentally designed to stabilize and interpret the noisy, periodic signals characteristic of quantum state measurements. Whether these signals originate from the optically detected magnetic resonance (ODMR) of an NV-diamond magnetometer, the spin-exchange relaxation-free (SERF) state of an atomic gyroscope, or the hyperfine transition of a cold-atom interferometer, the core technical challenge of **real-time noise filtering and high-fidelity state discrimination remains the same**. Our AI-native engine is architected to solve this general problem at the signal level, making it a truly modular and scalable solution for the entire suite of emerging quantum PNT sensors that the DoD is pursuing, from CubeSats to hypersonic weapons.

**OUSD(R&E) CRITICAL TECHNOLOGY AREAS (CTA) and/or COMMERCIAL SPACE STRATEGY MISSION AREA**

Our main CTA is **Space Technology.**
Our secondary CTA is **Trusted AI and Autonomy**.
Our main Commercial Space Strategy Mission Area is **Positioning, Navigation, and Timing (PNT)**

**TRANSITION PLAN**

Our transition plan is centered on a key principle: proactive integration and partnership. We will move in parallel on technology development and customer engagement to ensure our AI-Cognitive Engine is not just technically superior but is designed from day one to be seamlessly adopted by our USSF partners. Phase I delivers the core AI control engine; Phase II will deliver the complete, radiation-tolerant PNT stack.

**Integration Strategy and Timeline:**

- **Phase I (TRL 3):** Validate the AI control loop building block.
  - **Key Deliverable:** Develop a **Draft Interface Control Document (ICD)** for the AI Cognitive Engine. This document will define standard physical and logical interfaces (e.g., SpaceWire, MIL-STD-1553, Ethernet) and data protocols, ensuring the technology is designed from day one for seamless integration with existing spacecraft bus architectures and sensor front-ends, such as those being developed under the DIU TQS program.
  - Define integration requirements and Key Performance Parameters (KPPs) with transition partners.
  - Secure Letters of Support (LOS) and documented intent to partner from [redacted] and/or DIU.
- **Phase II (TRL 5):** Develop a brassboard prototype of the fused navigation engine. Integrate the controller with the NV-magnetometer and complementary sensors (MEMS IMU, CSAC) provided by hardware partners. Demonstrate fused PNT capability in a lab environment.
- **Phase III/Beyond (TRL 7+):** Transition to an on-orbit demonstration via the DIU TQS program or the DoD Space Test Program (STP), leading to insertion into a USSF Program of Record under [redacted].

[Customer engagement details omitted for publication]

**SUMMARY OF THE PROPOSED SOLUTION'S COMMERCIALIZATION & NON-DEFENSE COMMERCIAL POTENTIAL BUSINESS PLAN**

[Commercial strategy omitted for publication]

**Non-Defense Commercial Solution: A Space-Grade Edge AI Platform**

The commercial product is a standalone, radiation-tolerant Edge AI computer designed for the unique demands of space. While its "of a type" lineage includes other space-grade processors, our solution is novel in its specific focus on ultra-low-latency, hybrid AI/classical control for noisy sensor data, rather than general-purpose computing.

- **End-User Use Cases:**
    1. **Active & Assured PNT (Positioning, Navigation, and Timing):** This includes fusing IMU data with real-time terrain-matching from optical/LiDAR sensors (similar to a cruise missile's navigation) or performing opportunistic navigation by detecting and ranging signals from other LEO satellites. This provides a GPS-independent PNT capability that is highly sought after by commercial constellation operators concerned with resilience and signal spoofing.
    2. **Space Situational Awareness (SSA) & Sensor Cueing:** Fusing data from optical and RF sensors to autonomously detect, track, and characterize resident space objects. Our platform's <1ms control loop enables real-time "detect-to-cue" maneuvers, allowing a satellite to use a wide-view sensor to find an object and immediately slew a high-resolution sensor to it for inspection before it's lost.
    3. **Satellite Servicing & Rendezvous/Proximity Operations (RPO):** Providing the core "brain" for satellite servicing vehicles (like the Northrop Grumman MEV) or debris removal spacecraft. This requires fusing real-time data from LiDAR, machine vision, and IMUs to perform autonomous, high-precision docking maneuvers. A low-latency, high-reliability control loop is mission-critical for safe RPO, and our rad-tolerant platform is ideal for this high-stakes application.

**Competitive Landscape & Differentiation**

The space-grade computing market includes established players and agile startups. Our primary competitors are:

- **AMD (Xilinx):** Offers powerful, radiation-tolerant FPGAs (e.g., XQR Versal series) that are highly capable but represent a "build-your-own" solution requiring extensive customer NRE.
- **Ramon.Space:** Provides powerful AI/ML processors and software, but often at a higher power and cost point aimed at larger, more complex payloads.
- **Unibap (SpaceCloud®):** Offers an x86-based "cloud-in-space" architecture, focusing on software virtualization and a broader application ecosystem.

**Our Differentiator:** Our AI-Cognitive Engine is not a general-purpose space computer. It is a specialized **low-power sensor node processor.** We will win by focusing on a niche where we are superior:

1. **Power Efficiency:** Targeting **<12W** total system power, up to 50% lower than competing solutions for a similar processing task.
2. **Hybrid Architecture:** Our fusion of a µs-scale classical loop with an AI-adaptive loop is unique, providing the deterministic, low-latency control required for quantum sensors and other high-speed instruments that competitors cannot guarantee.
3. **Turnkey Solution:** We provide a complete hardware and firmware stack optimized for PNT, reducing customer integration time and NRE compared to buying a bare FPGA from a competitor.

[Commercial strategy and pricing omitted for publication]

**Technical Risks**

The primary technical risks and our mitigation strategies are:

1. **AI Generalization Gap:** The risk that our AI model, trained on simulated data, fails on real hardware. **Mitigation:** As detailed in our technical volume, we will use a dedicated Phase I task to develop realistic, physics-based noise models for data augmentation, ensuring the model is robust against real-world conditions.
2. **Supply Chain for Rad-Tolerant Components:** Dependence on a single supplier (Microchip for the RT PolarFire) creates risk. **Mitigation:** We will establish a direct relationship with Microchip's aerospace division in Phase I and explore alternative rad-tolerant FPGA options from other vendors (e.g., AMD, Renesas) for future product revisions.
3. **Sensor Integration Complexity:** Fusing multiple disparate sensors (IMU, CSAC, magnetometer) in a high-vibration, high-thermal-delta environment is non-trivial. **Mitigation:** Our Phase II plan includes extensive environmental testing (TVAC, Vibe) of the integrated brassboard prototype to de-risk these challenges early.

**RELATED WORK (DUAL-USE)**

**Similar previous and/or parallel work:**

**Significant Activities Directly Related to the Proposed Effort by our Team:**

[Prior customer work and contact details omitted for publication]

**Similar Previous & Parallel Work Not Conducted by Our Team:**

| Domain | Representative Work & Completion | Relevance / Interface with Our Project |
|---|---|---|
| **NV-Diamond Physics** | Faraday-effect non-destructive readout (Walsworth Group, 2024) | Higher-SNR readout would feed directly into our AI loop, improving sensitivity. |
| **Quantum-Control Theory** | Cyclic Geodesic Driving for robust sensing (Awschalom Group , 2024) | Our adaptive AI loop tackles the same noise problem but learns online, complementing pre-computed pulse protocols. |
| **Industrial Dual-Use** | 3-D current-density imaging in ICs with NV sensors (Wrachtrup Group , 2024) | Validates NV technology's commercial value, strengthening our dual-use transition story. |
| **Space-Grade AI Hardware** | AMD XQR Versal, Ramon.Space NuPod, Unibap iX-Series (2024–25) | These are general-purpose, high-power AI computers; our low-SWaP, sensor-specific controller fills the capability gap. |

**Relevant Government Initiatives Guiding Planned Coordination:**

- **Defense Innovation Unit (DIU) - Transition of Quantum Sensing (TQS):** Launched in 2024, the TQS program aims to rapidly demonstrate the military utility of quantum sensors for PNT. Its partnership with the Air Force to fly a quantum sensor on the X-37B signals an unambiguous, top-level commitment. Our project is perfectly aligned, providing the critical autonomous control technology necessary to evolve these DIU-prototyped sensors into robust, operational systems for the warfighter. We are not duplicating their sensor effort; we are providing a necessary component for its ultimate success.
- **DoD Space Test Program (STP):** Managed by the U.S. Space Force, STP is the primary provider of spaceflight opportunities for the DoD S&T community and the logical transition path for a flight demonstration of our technology after Phase II. Our system is designed to fit the STP mission profile for a CubeSat or hosted payload, demonstrating our long-term strategic vision and understanding of the DoD's technology transition pipeline.
- **DARPA & AFRL Initiatives:** DARPA's public investments in AI, quantum science, and resilient systems (e.g., the Robust Quantum Sensors - RoQS program) have laid the groundwork for this effort. Our project represents a novel synthesis of these key areas into a single, tangible capability. We are not aware of any current program developing a specialized, AI-based, real-time control system for in-space quantum sensors. Therefore, our work constitutes a new and promising solution to the persistent DoD challenge of achieving assured PNT.

**NON-PROPRIETARY WORK PLAN OUTLINE AND PHASE I OBJECTIVES**

**Scope:**

This Phase I effort will demonstrate the technical feasibility of a radiation-tolerant, hybrid AI control loop which is a core building block for a future, fully integrated PNT solution. The scope is focused on validating this key component through high-fidelity simulation and analysis against mission-relevant performance and SWaP-C targets. Full sensor integration with MEMS IMUs, CSACs, and star trackers is planned for Phase II.

**Major Objectives:**

**Objective 1 (Technical Feasibility):** To **Demonstrate**, via high-fidelity, noise-inclusive simulation, that the Hybrid AI Control Loop can suppress modeled on-orbit magnetic and thermal noise sources by at least an order of magnitude compared to a benchmark classical PID controller, resulting in a simulated heading error reduction of over 50%.

**Objective 2 (SWaP-C Validation):** To **Certify**, through the use of vendor-provided FPGA power estimation tools and static timing analysis, that the proposed architecture, when implemented on the RT PolarFire SoC, meets the stringent SWaP-C targets for a 12U CubeSat PNT payload, including a total power budget under 12W and all control loop latency targets (<150 µs classical, <1 ms AI).

**Objective 3 (Transition Pathway): Letters of Support (LOS) and documented intent to partner** from [redacted] and/or DIU TQS, that validates the defense need and defines the specific, quantitative Key Performance Parameters (KPPs) for a Phase II brassboard prototype and on-orbit demonstration.

[Contract certification omitted for publication]

| DURATION | DELIVERABLE TITLE | DELIVERABLE DESCRIPTION | SUCCESS MARKERS / OBJECTIVES | PERFORMER |
|---|---|---|---|---|
| Day 1-45 | Task 1: System Modeling & Requirements Definition | Task establishes the foundational physics-based model of the NV-diamond magnetometer and system architecture. We will generate the synthetic datasets for AI training and the project kickoff meeting. | • Validated sensor model and synthetic data generator complete.<br>• Kickoff meeting completed with USSF stakeholders (Day <30). | Jason Phillip |
| Day 30-75 | Task 2: AI Controller Development & Simulation | Using the validated models from Task 1, we will design, train, and quantize (INT8) the CNN controller. We will then conduct end-to-end simulations to benchmark the performance of our hybrid loop against a classical PID controller. | • Trained CNN achieves >99% state classification accuracy on test data.<br>• **(Supports Obj. 1)** Simulation demonstrates noise suppression >10x and heading error reduction >50% compared to the classical PID baseline. | FPGA & Embedded Systems Engineer |
| Day 45-90 | Task 3: FPGA Implementation & SWaP-C Analysis | We will implement the complete hybrid control loop architecture targeting the Microchip RT PolarFire SoC. We will use vendor tools to perform static timing analysis and generate accurate power estimates for the target hardware. | • **(Supports Obj. 2)** Static timing analysis confirms all control loop latency targets are met (<150µs classical, <1ms AI).<br>• **(Supports Obj. 2)** FPGA power estimation certifies total PNT payload power is <12W. | Power Electronics & Integration Engineer |
| Day 1-90 | Task 4: Stakeholder Engagement & Transition Planning | This task runs in parallel throughout the project. We will engage with transition partners at [redacted], DIU, and relevant primes to refine Phase II requirements and develop a detailed Concept of Operations (CONOPS). | • **(Supports Obj. 3)** Documented technical requirements and KPPs for a Phase II prototype, co-developed with a USSF stakeholder.<br>• A detailed CONOPS for an on-orbit demonstration is drafted. | System Architect & Integration Lead |
| Day 75-90 | Task 5: Final Reporting & Phase II Preparation | This task synthesizes all technical and transition activities into the required contract deliverables. We will draft the Phase II Customer Memorandum and secure letters of support based on the validated results from Tasks 2 & 3. | • Delivery of required reports, including Final Technical Report and Business Plan (Day <75).<br>• **(Supports Obj. 3)** A signed Letter of Support (LOS) from a relevant USSF stakeholder (e.g., [redacted]/CG-SYD) is secured for our Phase II effort. | Jason Phillip |

[Certification content omitted for publication]

**FOREIGN PERSONS**

[Personnel and certification content omitted for publication]

**GLOSSARY**

| Acronym | Definition |
|---|---|
| ADC | Analog-to-Digital Converter |
| ADCS | Attitude Determination and Control System |
| AFRL/RV | Air Force Research Laboratory, Space Vehicles Directorate |
| AI | Artificial Intelligence |
| CAGE | Commercial and Government Entity |
| CG-SYD | PNT Systems Directorate |
| CNN | Convolutional Neural Network |
| CONOPS | Concept of Operations |
| CSAC | Chip-Scale Atomic Clock |
| CSO | Commercial Solutions Opening |
| CTA | Critical Technology Areas |
| DAF | Department of the Air Force |
| DIU | Defense Innovation Unit |
| DoD | Department of Defense |
| DSIP | Defense SBIR/STTR Innovation Portal |
| DSP | Digital Signal Processing |
| FPGA | Field-Programmable Gate Array |
| GNC | Guidance, Navigation, and Control |
| GPS | Global Positioning System |
| ICD | Interface Control Document |
| IMU | Inertial Measurement Unit |
| ISR | Intelligence, Surveillance, and Reconnaissance |
| ITAR | International Traffic in Arms Regulations |
| KPPs | Key Performance Parameters |
| LEO | Low Earth Orbit |
| LOS | Letters of Support |
| MEMS | Micro-Electro-Mechanical Systems |
| MEV | Mission Extension Vehicle |
| NRE | Non-Recurring Engineering |
| NV | Nitrogen-Vacancy |
| ODMR | Optically Detected Magnetic Resonance |
| OUSD(R&E) | Office of the Under Secretary of Defense for Research and Engineering |
| PID | Proportional-Integral-Derivative |
| PLL | Phase-Locked Loop |
| PNT | Positioning, Navigation, and Timing |
| POC | Point of Contact |
| POW | Percentage of Work |
| pLEO | proliferated Low Earth Orbit |
| QML | Qualified Manufacturers List |
| R&D | Research and Development |
| R-GPS | Resilient Global Positioning System |
| RPO | Rendezvous and Proximity Operations |
| SBC | Small Business Concern |
| SBIR | Small Business Innovation Research |
| SEU | Single-Event Upset |
| SoC | System-on-Chip |
| SRAM | Static Random-Access Memory |

| Acronym | Definition |
|---|---|
| SSA | Space Situational Awareness |
| [redacted] | Space Systems Command |
| STP | Space Test Program |
| STTR | Small Business Technology Transfer |
| SWaP | Size, Weight, and Power |
| SWaP-C | Size, Weight, Power, and Cost |
| TMR | Triple Module Redundancy |
| TQS | Transition of Quantum Sensing |
| TRI | Topic Release Instructions |
| TRL | Technology Readiness Level |
| TVAC | Thermal Vacuum |
| UEI | Unique Entity ID |
| USAF | United States Air Force |
| USSF | United States Space Force |